

Student Internship and Job Portal with Matching

Functional Requirements:

- 1. User Registration & Authentication:** Students and admins must be able to register, log in, and authenticate.
- 2. Student Profile Management:** Students can create, update, and manage profiles including skills, education, and experience.
- 3. Resume Builder:** The system allows students to generate and download resumes based on profile data.
- 4. Job Posting Management (Admin):** Admins can create, edit, delete, and publish internship/job postings.
- 5. Job Search & Filtering:** Students can search jobs using filters such as skills, location, and job type (using LINQ).
- 6. Job Matching System:** The system automatically matches students to jobs based on their skills and profile data.
- 7. Application Submission:** Students can apply for jobs and upload resumes.
- 8. Application Tracking:** Students can track application status (e.g., pending, accepted, rejected).
- 9. Notifications System:** Students receive notifications for new matches, application updates (implemented using delegates).
- 10. Admin Dashboard & Reporting:** Admins can view analytics such as number of applications, active jobs, and user activity.

Non-Functional Requirements:

- 1. Performance:** The system should handle multiple users simultaneously within minimal time (< 500ms).
- 2. Scalability:** The system must support increasing numbers of users, job postings, and applications without performance degradation.
- 3. Security:** Sensitive data (passwords, personal info) must be encrypted and protected against unauthorized access.
- 4. Usability:** The interface (web and desktop) should be intuitive.

5. Reliability & Availability: The system should ensure high uptime (e.g., 99%) and maintain data access via the shared database.

Scenario:

Scenario 1: The Student

- 1. Registration:** The student opens the application, clicks "Register", enters "name_surname" and a password.
- 2. Profile Setup:** After login, the student goes to the "My Profile" tab and enters their skills: "C#, SQL, JavaScript".
- 3. Discovery:** The student switches to the "Browse Jobs" tab and types "SQL" in the search box.
- 4. Application:** The student finds a "Junior Developer" position in "Istanbul" and clicks "Apply". The status is now "Pending".
- 5. Notification:** Later, when the student logs in (or refreshes), they receive a popup notification saying their application for "Junior Developer" has been "Accepted".

Scenario 2: The Admin

- 1. Login:** The admin logs in.
- 2. Posting a Job:** The admin goes to "Manage Jobs", enters details for a "Full Stack Intern" role, and clicks "Add Job".
- 3. System Feedback:** A notification (via Delegate) confirms the job is live.
- 4. Reviewing Applications:** The admin goes to "Manage Applications", sees "name_surname" applied for the "Junior Developer" role.
- 5. Decision:** The admin clicks "Accept". The system updates the database and prepares the notification for the student.
- 6. Reporting:** The admin checks the "Reports" tab to see that the total number of "Accepted" applications has increased.